

7.2.5. Precision and Recall & F-score Example

Q: multiple answer Question

(a) (b) (c) (d)
✓ ✓ ✗ ✓

(a) choose (a)

(b) choose (a)(b)(c)(d)

(c) choose (a)(b)(d)

Calculate precision Recall F-score

Solution (a)

1	0
2	1

$$\text{precision} = \frac{1}{1+0} = 100\%$$

$$\text{Recall} = \frac{1}{1+2} = 33.3\%$$

$$F = \frac{2 \times 1 \times 0.3333}{1 + 0.3333} = 50\%$$

(b)

3	1
0	0

$$\text{pre} = \frac{3}{3+1} = 75\%$$

$$\text{Recall} = \frac{3}{3+0} = 100\%$$

$$F = \frac{2 \times 0.75 \times 1}{1.75} = 85.7\%$$

(c)

3	0
0	1

$$pre = \frac{3}{3+0} = 100\%$$

$$recall = \frac{3}{3+0} = 100\%$$

$$F = \frac{2 \times 1 \times 1}{1+1} = 1$$